

SALTDRAW · PRECIS

SaltDraw design precis

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DOCUMENT CONTROL

Document ID	SDG-PRC-001	Version	0.3
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/saltdraw		
Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-03	Amish Chadha	TRL 3; numbers from SDG-CAL-001 and the constructable design (SDG-DDR-002); bund roller and ballast added, windrow board removed
0.2	2026-10-03	Amish Chadha	TRL 2 precis; how it works, components, key design choices, first-order numbers, safety and concept media (SDG-DDR-001)
0.1	2026-09-30	Amish Chadha	Initial scaffold

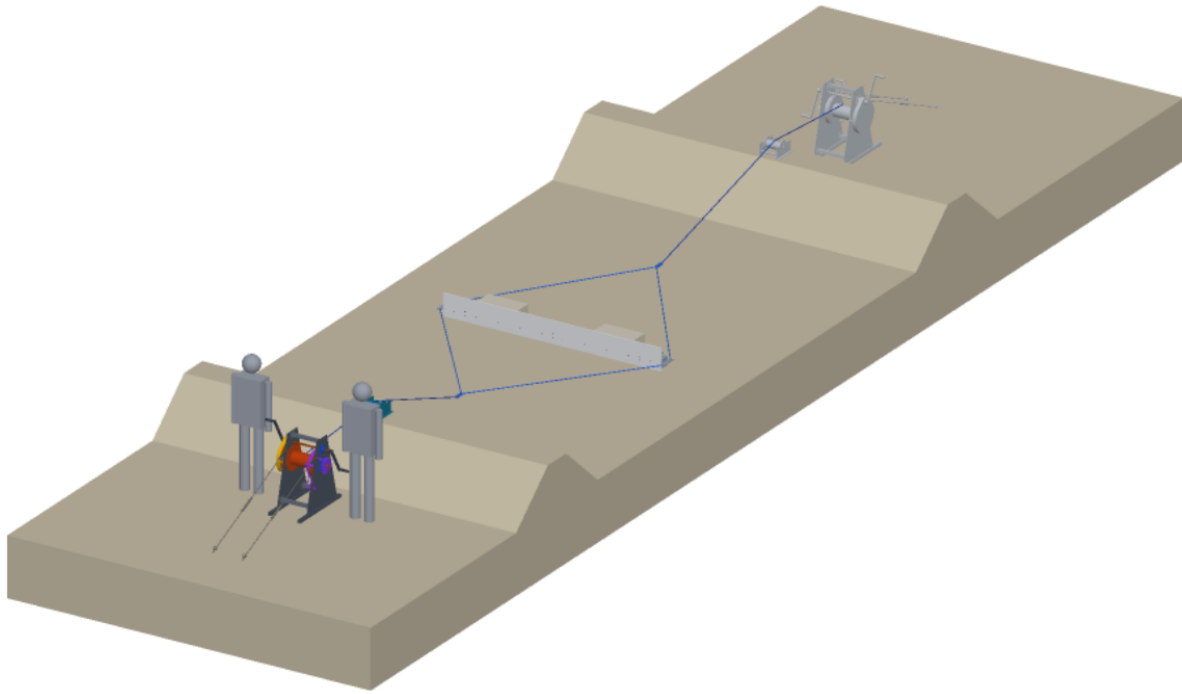
SaltDrag design precis

Drags a rake across the salt pan with hand capstans on the bunds so workers stay out of the brine.

SaltDrag

github.com/BoujeeEnjinia1701/saltdrag

CONCEPT, NOT FOR FABRICATION



Seen from the front right and above, 24 deg elevation. Grey: bunds and ground, far-side capstan and roller, two 1.75 m people at the cranks for scale

Figure 1. The working layout on a pan shortened to 7 m for the picture: the near capstan and bund roller in colour, the far-side set in grey, two 1.75 m people at the cranks for scale (concept media, SDG-DWG-010).

■ How it works

Two people winding a hand capstan on dry ground behind one bund draw a 3 m rake across the pan by a rope that runs over a roller on the bund crest. The rake's plastic blade rides 10 mm above the floor on three skids and pushes the loose salt crystals ahead of it into a heap of about 230 kg. At the near bund the steepening rope starts to lift the rake about 3.3 m from the bund toe, so the heap is left there as a windrow. A second capstan on the far bund holds a light back tension while the rake is pulled in, then winds it back empty. Both capstans then move 3 m along their bunds, and the return pull brings the rake onto the next strip. Nobody stands in the brine or walks on the pan at any stage.

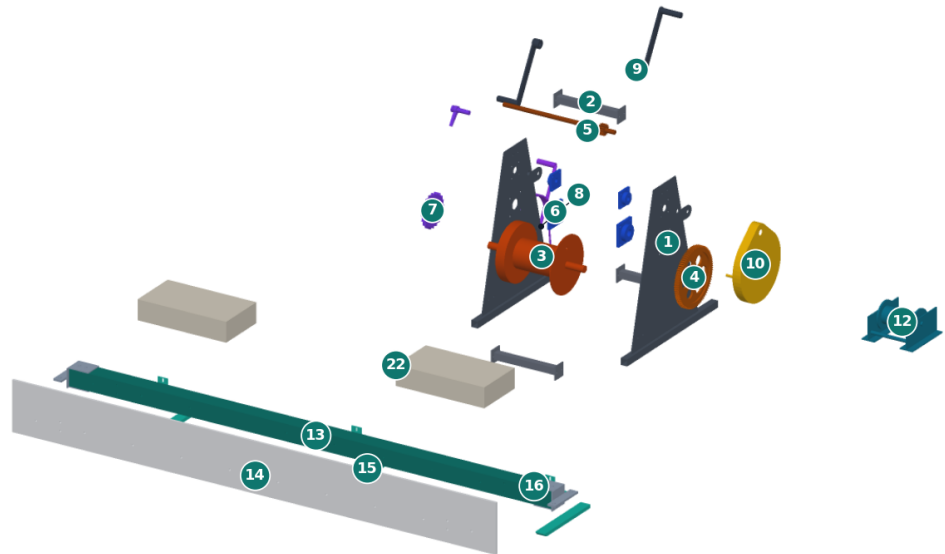
Each capstan is a geared hand winch: two cranks drive a 14-tooth pinion, which turns an 84-tooth gear on the drum shaft (6 to 1). A pawl on a ratchet wheel stops the drum running back whenever the cranks are let go, and a band brake lets the far capstan pay out rope under control. Two chains from ears on the side frames, at rope height, hold the capstan to two screw-in ground anchors behind it.

SaltDrag: exploded view

github.com/BoujeeEnjinia1701/saltdrag

CONCEPT, NOT FOR FABRICATION

- 1 Capstan side frames
- 2 Cross tubes
- 3 Drum
- 4 Spur gear
- 5 Crank shaft and pinion
- 6 Flange bearings
- 7 Ratchet and pawl
- 8 Brake band and lever
- 9 Crank handles
- 10 Gear guard
- 12 Bund roller
- 13 Rake beam
- 14 Rake blade
- 15 Skids
- 16 End caps
- 22 Ballast bags



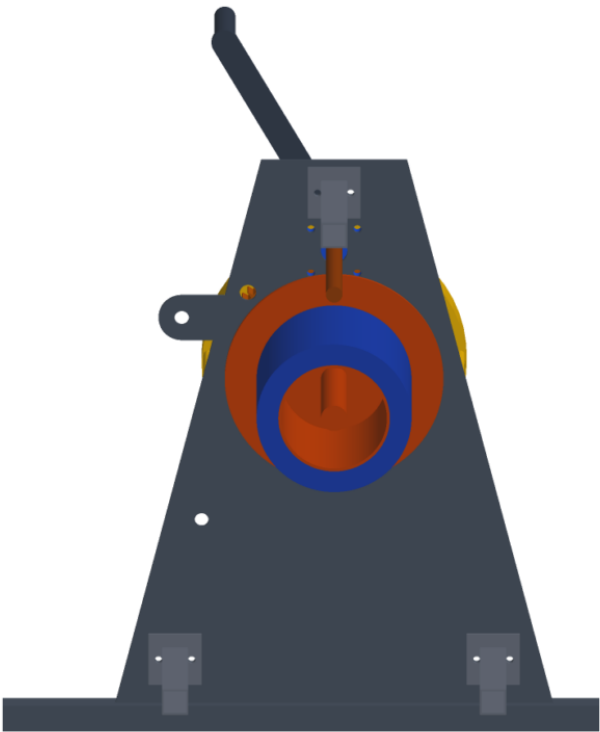
One capstan (two are built), the rake (front) and a bund roller (right), pulled apart; seen from the front right and above, 24 deg elevation; numbers match bom/bom.csv

Figure 2. One capstan, the rake and a bund roller pulled apart; numbers match bom/bom.csv.

SaltDrag: capstan cutaway

github.com/BoujeeEnjinia1701/saltdrag

CONCEPT, NOT FOR FABRICATION



Capstan turned so the drum axis runs away from the viewer; the near half (ratchet and brake side) removed; seen from the front and above, 18 deg elevation; rope shown three layers deep

Figure 3. Capstan with its ratchet and brake side removed, showing the drum, flanges and three layers of rope.

■ Components

Table 1. Main components (two capstans, two rollers and two ropes in a set).

BOM	Component	Role
1, 2	Side frames and cross tubes	Two 6 mm galvanised cheeks on box-section runners, bolted together by three cross tubes; the runners let the capstan be slid along the bund
3	Drum	168 mm barrel between 330 mm flanges on a 40 mm shaft; holds 58 m of 12 mm rope; the left flange carries the brake rim
4, 5, 9	Gear, crank shaft and pinion, crank handles	Module 4 spur gears, 6 to 1; two removable 330 mm cranks set 180 degrees apart
6	Flange bearings	Four-bolt bearings with stainless inserts in thermoplastic housings
7, 8	Ratchet and pawl; brake band and lever	The pawl holds the load; the band brake controls paying out
10	Gear guard	Yellow sheet-steel cup over the gear mesh
11	Ground anchors, chains and turnbuckles	Two screw-in anchors per capstan, chained to the cheek ears
12	Bund roller saddle	UHMW-PE roller on a small galvanised saddle astride the crest, so the rope never cuts the bund
13 to 16	Rake beam, blade, skids, end caps	3.0 m FRP box beam with a 300 mm UHMW-PE blade; skids set the blade height; stainless end caps carry the bridle eyes
17, 18	Bridles and pull ropes	Two-leg polyester bridles to a swivel; 45 m of 12 mm polyester double braid per side
20, 22	Rope dampers and ballast bags	A weighted bag on each loaded rope; two salt-filled bags keep the blade down near the bund
21	Maintenance kit	Rinse water, grease and spares

■ Key design choices

All were decided on 2026-10-03 under Amish's pre-approval of this batch (SDG-DDR-001 and SDG-DDR-002; register SDG-DEC-001).

- **Drum winch, not a warping capstan.** The drum stores the rope, so nobody has to tail a loaded rope by hand.
- **Capstan behind the bund, roller on the crest.** A 0.4 m crest is too narrow for a capstan and two people cranking, and the roller keeps the rope off the bund.
- **Hand power, single spur stage.** No engine or motor, in keeping with the Agariya registration conditions.
- **Plastic and FRP in the pan, galvanised steel on dry ground.** Nothing that rusts sits in the brine.
- **One-piece 3 m rake with ballast.** The rake width keeps the design drag at the rated 3 kN; salt-filled bags keep the blade down as the rope steepens near the bund.

■ First-order numbers

Table 2. Key figures (estimates from SDG-CAL-001).

Quantity	Value	Basis
Full heap in front of the blade	0.19 m ³ , about 231 kg	300 mm blade, 35 degree repose, 1,200 kg/m ³ loose salt
Design drag on the rake	2.75 kN	Heap sliding 1.59 kN, layer loosening 0.90 kN (assumed 300 N/m), skids 0.26 kN
Rope pull at the drum	3.0 kN	Drag, roller loss and 200 N back tension
Crank force, two people	89 N each	6 to 1, 330 mm cranks, 94 % gear and bearing efficiency, rope on its third layer
Pull with two people at 120 N	4.0 kN (3.7 kN on a full drum)	R3 asks for 3 kN
Rope speed and work rate	3.1 m/min at 30 crank turns a minute; 84 W each	Light to moderate work, early hours in the hot season
One strip	23 min: pull 9.5, return 5.7, move 8	30 m pan
Gathering rate	About 600 kg/h, 150 kg per worker-hour (crew of four)	Baseline for R5 still to be measured
Heaviest single part	23.8 kg (ballast bag, filled on site); drum 21.3 kg	Model masses
Parts cost of the set	USD 2,803	Value-engineering target USD 4,000 (USD 1,197 under)

SaltDrag: work in one 30 m pass with a full heap, kJ (SDG-CAL-001 estimates)

github.com/BoujeeEnjinia1701/saltdrag

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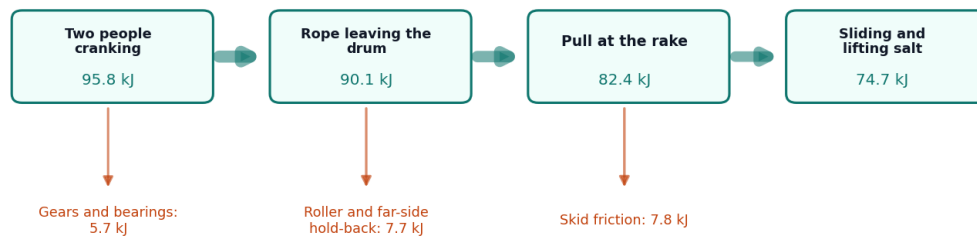


Figure 4. Where the cranking work goes in one 30 m pass with a full heap (estimates).

■ Patent design-arounds

From the preliminary patent, trademark and prior-art screen (not legal advice):

- No motorised or track-mounted auger harvester.
- Build on expired prior art such as the CA1243940A salt harvester; review CN102476810A (salt-pan rake) claims before release.

■ Shared blocks

- Hand capstan common block, shared with SiltHaul: this capstan is the first candidate for the common block (SDG-DDR-001). SiltHaul is not changed by this work.
- CalRig proof-load for rope, capstan and anchor load tests.

■ Safety

Safety: A loaded rope stores energy and can whip back if it breaks, slips off the drum or loses its anchor. The rope is low-stretch polyester with a working-load factor of 7.7 at the rated pull, each loaded rope carries a weighted damper bag at mid-span, and nobody stands in line with a loaded rope or on the bund within 2 m of it.

Safety: Cranks and gears can strike or trap hands. The pawl is always engaged while winding, so a released crank cannot spin back; the gears run inside a yellow guard; the crank handles are taken off before any rope is paid out under load, and paying out is done only on the brake.

Safety: A ground anchor that pulls out lets the capstan lurch toward the pan. Each anchor is proof-loaded on site to 1.5 times its share of the rated pull before first use, and the chains are fixed at rope height so the capstan cannot tip.

Safety: Heat is a serious hazard in the harvest season; plan work for cooler hours and provide shade and water. Parts of up to 24 kg are lifted and carried; lift in pairs.

This design is published as an open engineering reference, never as certified lifting or rope equipment. It is not certified equipment.

■ Open questions

No design decisions are open (register SDG-DEC-001). Facts that only a site survey or real parts can settle, such as the drag per metre on a real crystal bed, the bund crest width and the soil behind the bunds, are listed in the register to be confirmed.